

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 20 (800 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 20 (800 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on Quartz substrate with a target center frequency of 2208.0 MHz and bandwidth of 292.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 20 (800 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: Quartz
- Package: Fan-Out WLP
- Design Tool: CST Studio Suite 2024
- Design Center: Osaka Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 20 (800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2208.0 MHz
Bandwidth (3 dB):	292.0 MHz
Insertion Loss Target:	≤ 3.5 dB
Return Loss Target:	≥ 10.0 dB
Out-of-Band Rejection:	≥ 49.0 dB
Isolation (Duplexer):	≥ 19.0 dB
Q Factor:	6578
Temperature Coefficient:	-24.2 ppm/°C
Die Size:	1.4 x 1.4 mm
Wafer Size:	6-inch
Number of Resonators:	8
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: Quartz with specified layer thicknesses.
3. Set design targets: center freq 2208.0 MHz, BW 292.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Quartz).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 6624 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.5 dB
- Return loss in passband shall be >= 10.0 dB
- Out-of-band rejection at +/- 438 MHz offset >= 49.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 6.7 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-12-30
Devices Tested: 41
Insertion Loss (meas): 3.17 dB
Return Loss (meas): 12.4 dB
Rejection (meas): 52.2 dB
Isolation (meas): 23.9 dB
Q Factor (meas): 6578
Group Delay Variation: 6.8 ns
Power Handling: +27 dBm
Die Yield: 91.2%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated excellent performance for Band 20 (800 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. Hans Weber	Date:	2025-01-06

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